

Test 5 — Answer Key

Each row gives the item number, the B.E.S.T. benchmark and difficulty, the correct answer, and a student-facing explanation.

#	Benchmark	Answer	Explanation
1	MA.6.NSO.1.1 moderate	-2 1/2, -1 3/4	A number is less than -1 1/2 when it lies to the left of -1 1/2 on the number line.
2	MA.6.NSO.1.3 moderate	-4 1/2, -6 1/2, 7	Compare how far each number is from 0, ignoring its sign.
3	MA.6.NSO.1.2 moderate	There is no money left on the card and no amount is owed.	In a balance context, 0 is the point between having credit and owing money.
4	MA.6.AR.2.3 moderate	15	Divide both sides by the coefficient to undo the multiplication.
5	MA.6.GR.2.1 easy	base 7 centimeters, height 4 centimeters → 14; base 9 centimeters, height 6 centimeters → 27; base 6 centimeters, height 10 centimeters → 30	Use one half times base times height for each triangle.
6	MA.6.NSO.4.1 moderate	18	Adding a negative number moves left on the number line; adding a positive number moves right.
7	MA.6.GR.1.3 easy	22	Add the four side lengths, or use 2 times the sum of the length and width.
8	MA.6.AR.1.2 easy	36.0	The boundary value is 36, the number the inequality compares x to.
9	MA.6.NSO.2.2 moderate	1/3	Multiply numerator by numerator and denominator by denominator, then simplify.
10	MA.6.AR.1.4 easy	$2(x + 2) \rightarrow 2x + 4$; $2x + 7x \rightarrow 9x$; $2x + 2 - 2 \rightarrow 2x$	Use the distributive property and combine like terms.
11	MA.6.NSO.2.1 moderate	20.655	Multiply as with whole numbers, then place the decimal point using the total number of decimal places in the factors.
12	MA.6.AR.2.2 moderate	33	Subtract the same number from both sides to undo the addition.
13	MA.6.DP.1.3 moderate	16	The median is shown by the vertical line inside the box.
14	MA.6.AR.3.3 moderate	20	Each row multiplies the first ratio by the same number, here 5.
15	MA.6.NSO.2.3 moderate	\$38.83	Multiply the price by the number of items, then add the one-time fee.

#	Benchmark	Answer	Explanation
16	MA.6.DP.1.1 moderate	What are the heights, in inches, of the students in the band?	A statistical question expects different answers from different people, so it produces a set of numerical data.
17	MA.6.NSO.3.4 moderate	$2 \times 3^2 \times 7$	Break the number into factors until every factor is prime, then use exponents.
18	MA.6.NSO.3.3 moderate	1296	6^4 means 6 multiplied by itself 4 times, not 6×4 .
19	MA.6.NSO.3.5 moderate	15%	Write the fraction as a decimal, then multiply by 100.
20	MA.6.AR.2.4 moderate	3.65	Subtract the known addend from the total.
21	MA.6.DP.1.4 moderate	11	Count one dot for each student at 4 and to the right of 4.
22	MA.6.DP.1.2 moderate	23	Add all the values, then divide by how many values there are.
23	MA.6.AR.3.2 moderate	\$6.00	Divide the total cost by the number of pounds to get the cost for one pound.
24	MA.6.AR.3.5 moderate	432 inches	Each foot is 12 inches, so multiply the number of feet by 12.
25	MA.6.AR.1.1 easy	a number x decreased by 11 $\rightarrow x - 11$; the sum of a number x and 11, divided by 2 $\rightarrow (x + 11)/2$; the product of 11 and a number $x \rightarrow 11x$	Translate each phrase one operation at a time.
26	MA.6.AR.3.4 moderate	48	40% means 40 out of every 100, so multiply the total by 40/100.
27	MA.6.NSO.1.4 moderate	-12, 11	A reading sets off the alarm when its absolute value is greater than 10.
28	MA.6.NSO.4.2 moderate	64	The product of two numbers with the same sign is positive; with different signs it is negative.
29	MA.6.GR.2.2 easy	108, 12	Decompose the figure: whole rectangle minus the missing corner.
30	MA.6.DP.1.6 moderate	21 $\frac{1}{3}$	Add the new value to the total and divide by the new number of values.
31	MA.6.GR.2.4 moderate	52	Add the areas of all six rectangles in the net; opposite faces match.
32	MA.6.GR.1.1 easy	D (5, -2) $\rightarrow$ Quadrant IV; A (3, 4) $\rightarrow$ Quadrant I; C (-4, -3) $\rightarrow$ Quadrant III	The signs of the coordinates tell which quadrant a point is in.

#	Benchmark	Answer	Explanation
33	MA.6.NSO.3.1 moderate	6	List the factors of each number and choose the largest one they share.
34	MA.6.AR.1.3 easy	$6x + 4 \rightarrow 52$; $6x^2 \rightarrow 384$; $4 - x \rightarrow -4$	Substitute the value of x, then follow the order of operations.
35	MA.6.NSO.2.1 moderate	0.4×6 , $24 \div 10$, 1.2×2	Evaluate each expression and compare it to the target value.
36	MA.6.AR.3.1 moderate	9:7	Write bus students first, then divide both numbers by their greatest common factor.
37	MA.6.NSO.3.2 moderate	$6(2 + 5)$	The GCF of 12 and 30 is 6, and $12 = 6 \times 2$, $30 = 6 \times 5$.
38	MA.6.DP.1.5 moderate	2, 9, 20, 30, 37	Order the data, find the median, then find the medians of the lower and upper halves.